Ysbyty Gwynedd
Anaesthetics Department
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POST OPERATIVE

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POSTOPERATIVE NAUSEA AND VOMITING GUIDELINE

REVIEW BY: OCTOBER 2027

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Key Points

Indication:

- This guideline is designed to provide a comprehensive evidence based clinical recommendations on the management of PONV in adults and children.
- Prevention of PONV should be considered an integral aspect of anaesthesia, achieved through risk assessment.
- Specific advice can be obtained from the Acute Pain Service (APS) Monday – Friday (office hours) via pager 115.

INTRODUCTION

Nausea and vomiting are two of the most common adverse events in the postoperative period with an estimated incidence of 30% in the general surgical population and as high as 80% in high-risk cohorts (Apfel et al., 1999) This can be a highly distressing experience and is associated with significant patient dissatisfaction (Eberhart et al., 2002; Myles et al., 2000). In addition, the occurrence of postoperative nausea and vomiting (PONV) is also associated with a significantly longer stay in the postanaesthetic care unit (PACU) (Habib et al., 2006), unanticipated hospital admission (Fortier et al., 1998), and increased health care costs (Hill et al, 2000).

Although the side-effect profiles of equianalgesic doses of opioids are similar, individual patients may report more PONV with one particular opioid. In this situation changing to another opioid (for example morphine to fentanyl) is worth considering when other measures, such as the appropriate administration of antiemetics has failed.

It must be remembered that opioids are only one of the many factors that can influence the incidence of PONV. Other reasons include younger patient age, female gender, being a non-smoker, history of motion sickness or previous PONV, general anaesthesia (compared with

regional anaesthesia), some types of surgery, and the use of volatile agents and nitrous oxide (Gan et al., 2020; Schug et al, 2020).

Coadministration of other agents that enable good pain relief to be obtained with a lower dose of opioid ("opioid-sparing") may also lead to a reduction in the incidence of PONV and should always be considered (Gan et al, 2020). Opioids sparing with lower risk of PONV has followed coadministration of nonsteroidal anti-inflammatory drugs, ketamine, gabapentinoids nd clonidine (Schug et al., 2020)

A number of antiemetic drugs are available and they differ with regards to the site of action. Therefore, as with analgesic agents, the use of combination of antiemetic drugs that work of different receptor sites may be more effectives than a single drug. If one drug is ineffective, then the drug from another class should be added to the antiemetic regime.

ANTIEMETICS

There are a number of different classes of antiemetics drugs that act of various receptor sites involved in the emetic response – serotonin (5-hydroxytryptmine [5-HT], acting on the 5-HT_3 receptor), dopamine, acetylcholine (acting on the muscarinic receptors), histamine (H_1), and neurokinin (NK_1). Corticosteroids are also effective antiemetic agents.

5-HT_3 Receptor Antagonists

Antagonists at the 5-HT_3 receptor include ondansetron. They are among the most effective in the prevention and treatment of PONV, generally well tolerated with few significant side effects (Zabirowicz & Gan, 2019). Those that have been reported include headache, constipation, and dizziness, and may lead to prolongation of the QT interval in some patients.

Dopamine Receptor Antagonists

Metoclopramide is a dopamine receptor antagonist and used as an antiemetic as well as a prokinetic agent (that is, stimulates gastric motility and increases intestinal peristalsis). However, it has little is any antiemetic effect in the commonly used dose of 10mg (Zabirowicz & Gan, 2019). Extrapyramidal side effects including acute dystonic reactions can occur. Although the incidence is dose dependent, they may also occur after just a single dose in the same person.

Other dopamine receptor antagonist includes prochlorperazine.

H_1 - Receptor Antagonists

Cyclizine is also used as an antiemetic, but less well studied than the drugs listed earlier. It can lead to sedation. Extrapyramidal side effects can also occur.

Neurokinin-1 (NK-1) Receptor Antagonists

Aprepitant results in less vomiting compared with other antiemetics, but its efficacy against nausea is similar to other antiemetics (Zabirowicz & Gan, 2019). It is the first drug in this class to be developed and approved for use as it is non-sedating.

Anticholinergic Agents

Scopolamine (hyoscine) has also been used to treat nausea and vomiting (Zabirowitz and Gan, 2109). It is available in a transdermal patch as well as oral and parental formulations and is particularly effective for movement induced nausea and vomiting. It may be associated with significant anticholinergic side effects such as sedation, dry mouth, visual disturbances, and confusion and is not widely used in the management of PONV.

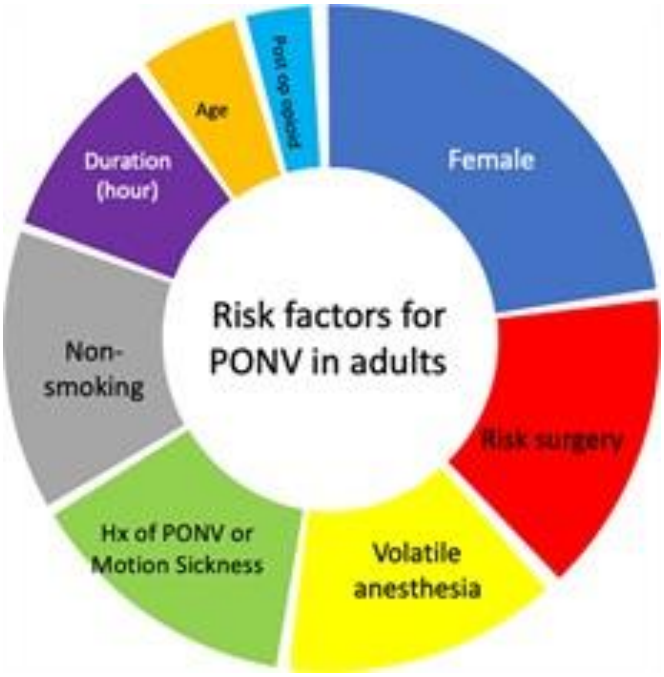
Corticosteroids

Dexamethasone is a very effective antiemetic, although its mechanism of action remains unclear. Most of the evidence to date suggests that administration of a single dose only will not increase the risk of postoperative infection (Gan et al. 2020). When given by injection to awake patients, some may complain of perineal burning and itching (Zabirowicz & Gan, 2019).

PONV RISK FACTORS

Intraoperative and postoperative risk factors in PONV in adults, the size of each segment is proportional to the offs of ratios of PONV associated with each risk factor (Apfel et al, 2012)

Figure 1. The PONV risk factor summary



Patient Risk Assessment for PONV

PONV risk factors should be used for risk assessment and to guide PONV management (Gan et al., 2013). PONV risk scores have been shown to reduce the rate of PONV at an institutional level and can be used to inform and guide therapy (Pierre et al., 2002,2004; Apfel et al., 2004). A commonly used risk scores for inpatients undergoing anaesthesia is the Apfel score (Apfel et al., 1999, Koivuranta et al, 19197). Figure 2. Illustrates the Apfel simplified risk score is based on 4 predictors: female sex, history of PONV and/or motion sickness, non-smoking status, and use of postoperative opioids. The incidence of PONV with the presence of 0, 1, 2, 3, and 4 risk factors is approximately 10%, 20%, 40%, 60%, and 80%, respectively (Apfel, 2014). The panel classifies patients with 0–1, 2, or 3-plus risk factor into “low,” “medium,” and “high” risk categories, respectively. Risk scores represent an objective approach to predict the incidence of PONV or PDNV, with sensitivity and specificity of between 65% and 70%, and should be utilized as a modifier for prophylaxis.

Figure 2. Risk score for PONV in adults

Risk Factors	Points	Percentage
No risk factors	0	10%
Female Gender	1	20%
Non-smoker	1	40%
History of PONV and/or motion sickness	1	60%
Postoperative opioids	1	80%
Total of point 0-4		

Simplified risk score (Apfel et al, 2012) to predict the patient's risk for PONV 0,1,2,3,4, risk factors correspond to PONV risks of approximately 10%, 20 %, 40%, 60% and 80% respectively.

Recommended PONV prophylaxis (at the discretion of the anaesthetist)

Level of risk	Score	Drug Regime
Low	0-1 point (20% or less)	No prophylaxis
High	2-3 point (40-60%)	Dexamethasone 3.3mg IV **AND** Ondansetron 4mg IV
Very high	4 points (80%)	Dexamethasone 3.3mg IV **AND** Ondansetron 4mg IV **AND** Cyclizine 50mg IV **OR CONSIDER** Propofol TIVA and antiemetic

Consider dual antiemetic prophylaxis for low-risk patients undergoing surgery with increased potential for PONV or where surgical outcomes may be compromised by PONV.

Rescue treatment of PONV in the recovery ward

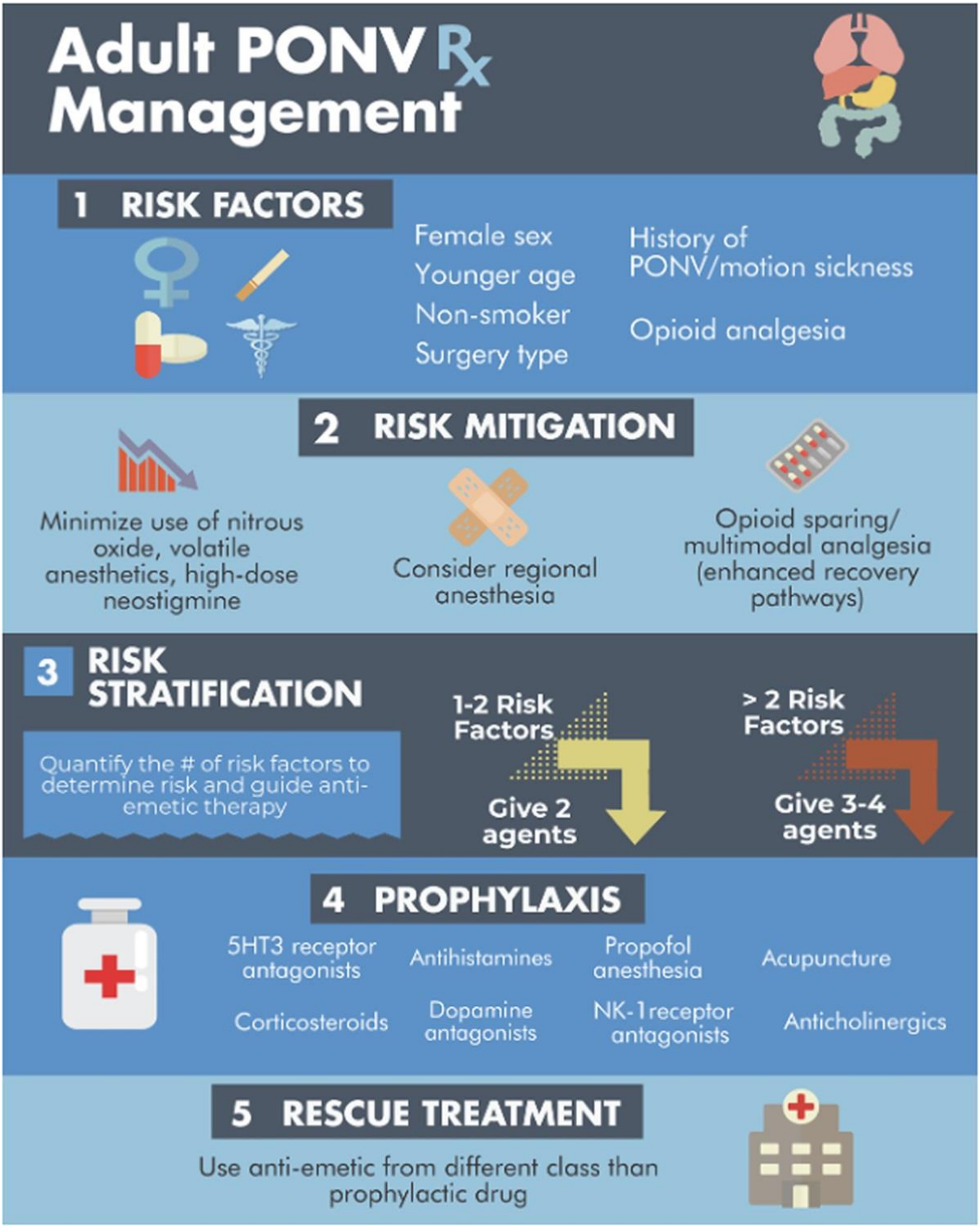
Check the drug chart and anaesthetic record before deciding on antiemetic treatment
Address factors such as pain, hypotension, hypoxia, hypothermia pyrexia and sepsis.

Antiemetics have and additive effect when used in combination. If a patient has been given ondansetron in theatre, use a different class of drug in theatre.

First line therapy	Ondansetron 4mg IV TDS if not given in the last 6 hours (otherwise give 2nd line therapy)
Second line therapy	Cyclizine 25-50mg IV TDS if not given in the last 8 hours (otherwise give 3rd line therapy)
Third line therapy	Metoclopramide 10mg IV TDS **or** Prochlorperazine 12.5mg IM TDS **or** Prochlorperazine 3-6mg Buccal BD

Figure 6 - Algorithm for PONV management in adults.

Summary of recommendations for PONV management in adults, including risk identification, stratified prophylaxis, and treatment of established postoperative nausea and vomiting. Note that 2 antiemetics are now recommended for PONV prophylaxis in patients with 1–2 risk factors. 5-HT$_3$ indicates 5-hydroxytryptamine 3; PONV, postoperative nausea and vomiting (Zabirowicz et al, 2019).



Assessment for PONV/POV Risk in Children

The risk factors for POV/PONV in children are different from those in adults (Eberhart et al., 2004; Thomas et al., 2007; Lee et al., 2016) and are summarized in Figure 3. Children are more at risk for PONV/POV when they are older than 3 years, subjected to certain surgeries—namely tonsillectomy and eye surgeries, or are post pubertal females (evidence B1). The other risk factors are summarized in the aforementioned figure and have been validated by Kranke et al (2007) by using the Postoperative Vomiting in Children (POVOC) score (Chau et al., 2017; Efune et al., 2018).

As previously proposed by Eberhart et al, (2004). POV risk in children can be predicted based on 4 criteria: duration of surgery >30 minutes; age >3 years; personal or first-degree relative history of POV/PONV; and strabismus surgery. Based on the presence of 0, 1, 2, 3, and 4 factors, the risk of POV was 9%, 10%, 30%, 55%, and 70%, respectively (Figure 5). This was subsequently verified by Kranke et al. 2007).

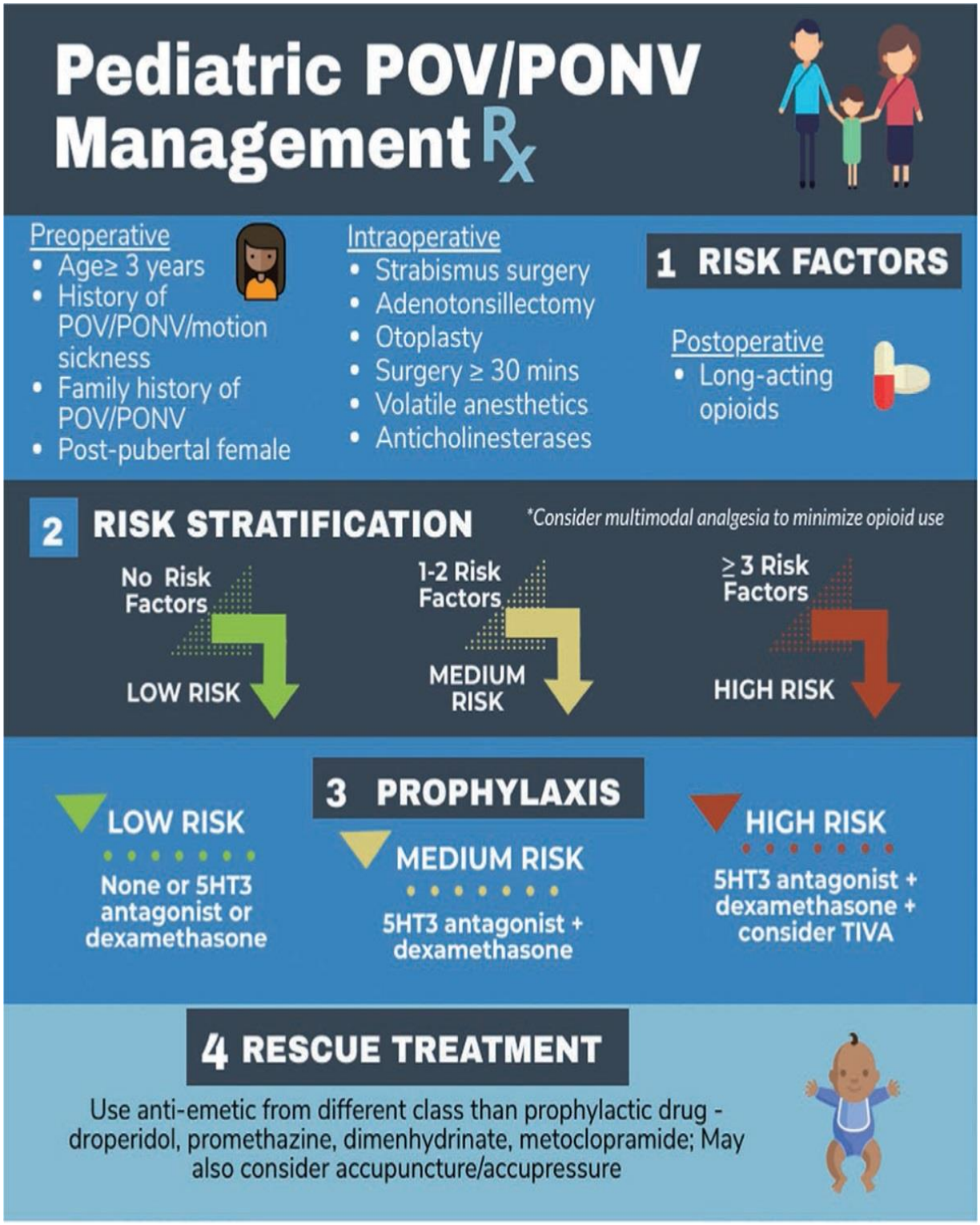
Figure 5. Risk score for POV in children

Simplified risk score from Eberhart et al (2004) to predict the risk for POV in children. 0, 1, 2, 3, or 4 risk factors correspond to POV risks of approximately 10%, 10%, 30%, 50%, or 70%, respectively. PONV indicates postoperative nausea and vomiting; POV, postoperative vomiting.

Risk Factors	Points	Percentage
Surgery ≥ 30 minutes	1	10%
Age ≥ 3 years	1	10%
Strabismus surgery	1	30%
History of POV or family history of PONV	1	50%
Total of point 0-4		

Figure 4. - Algorithm for POV/PONV management in children.

Summary of recommendations for POV/PONV management in children, including risk identification, risk-stratified prophylaxis, and treatment of established postoperative vomiting. 5-HT_3 indicates 5-hydroxytryptamine 3; PONV, postoperative nausea and vomiting; POV, postoperative vomiting; TIVA, total intravenous anaesthesia.



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